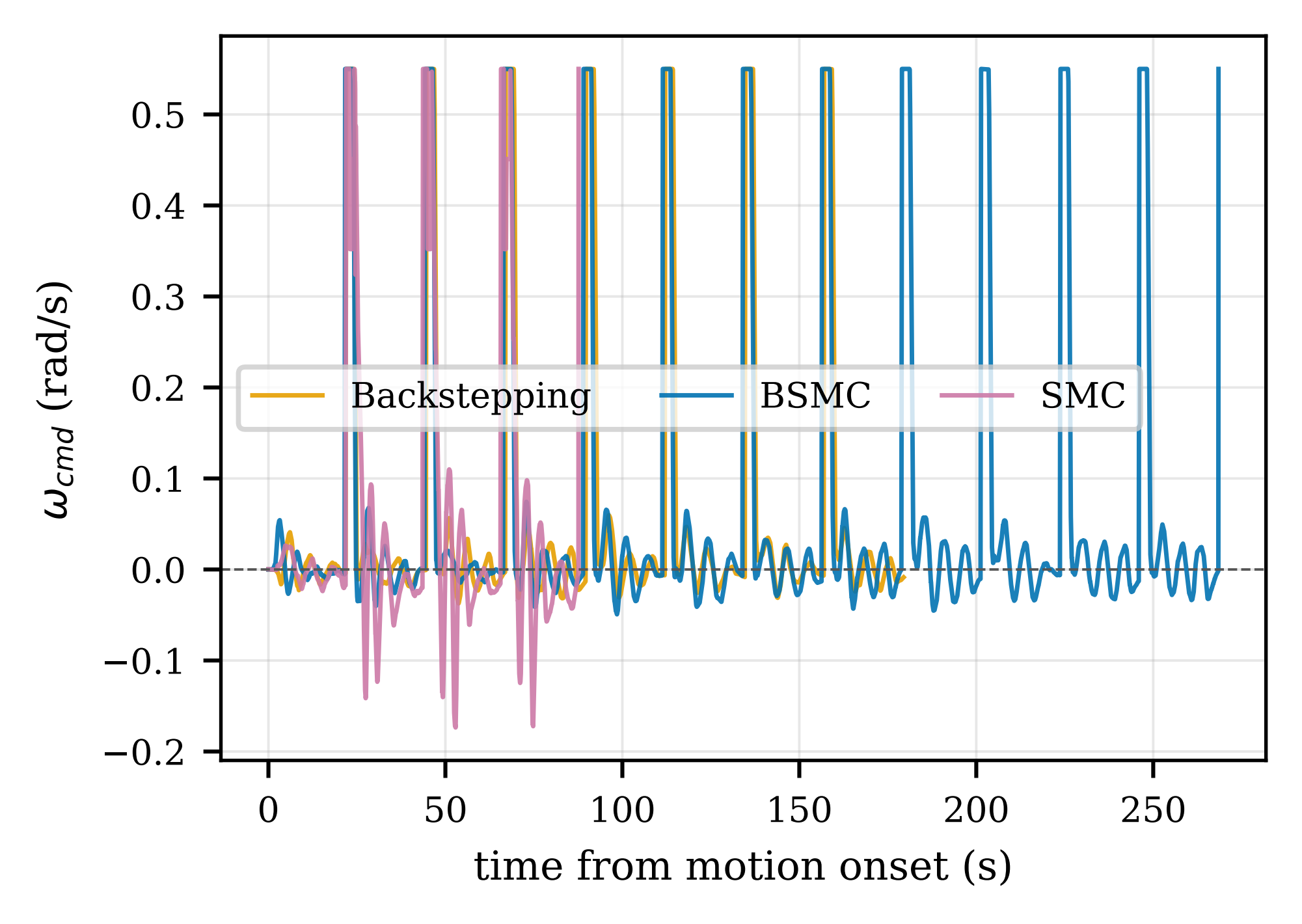
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
ω_{cmd} (rad/s)
Backstepping
BSMC
SMC
0
50
100
150
200
250
time from motion onset (s)